



Supervised Learning - Linear Classification

NYU K12 STEM Education: Machine Learning

Department of Electrical and Computer Engineering,
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Brooklyn, New York

- ▶ [Course Website](#)
- ▶ Instructors:



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Outline

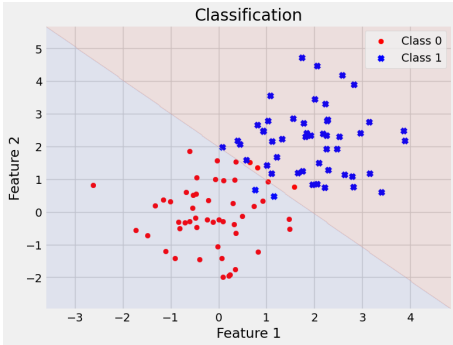
1. Linear Classification

2. Lab I

3. Multiclass Classification

4. Lab II

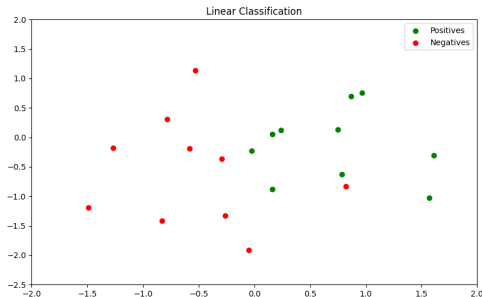
Classification vs. Regression



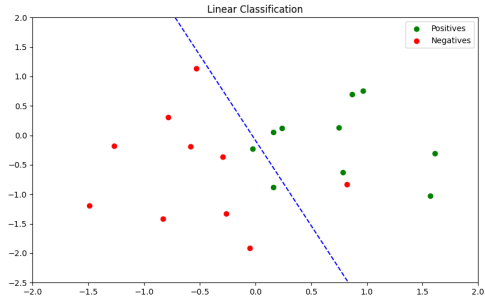
Classification

Given the dataset (x_i, y_i) for $i = 1, 2, \dots, N$, find a function $f(x)$ (model) so that it can predict the label $\hat{y}$ for some input x , even if it is not in the dataset, i.e. $\hat{y} = f(x)$

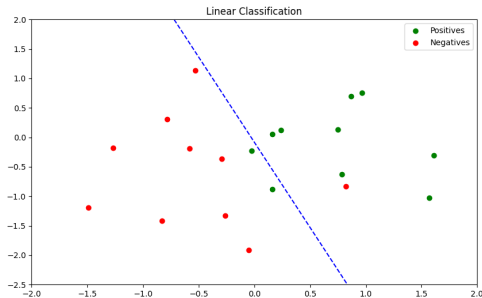
- Positive : $y = 1$
- Negative : $y = 0$



Decision Boundary



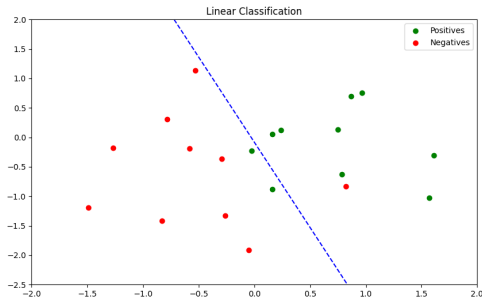
Decision Boundary



► Evaluation Metric:

$$\text{Accuracy} = \frac{\text{Number of correct prediction}}{\text{Total number of prediction}}$$

Decision Boundary

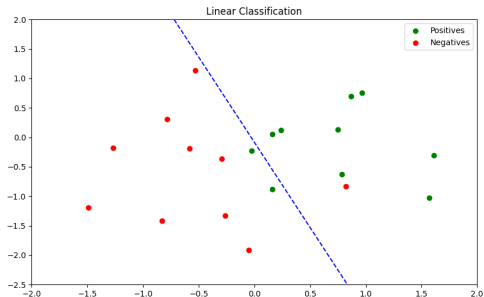


- Evaluation Metric:

$$\text{Accuracy} = \frac{\text{Number of correct prediction}}{\text{Total number of prediction}}$$

- What is the accuracy in this example?

Decision Boundary



$$\text{Accuracy} = \frac{\text{Number of correct prediction}}{\text{Total number of prediction}} = \frac{17}{20} = 0.85 = 85\%$$

Need for a new Model

- What would happen if we used the linear regression model:

$$\hat{y} = w_0 + w_1x$$

Need for a new Model

- ▶ What would happen if we used the linear regression model:

$$\hat{y} = w_0 + w_1x$$

- ▶ y is 0 or 1
- ▶ $\hat{y}$ will take any value between $-\infty$ and ∞

Need for a new Model

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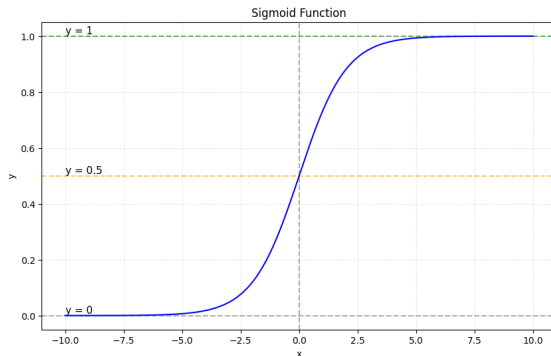
$$\hat{y} = w_0 + w_1x$$

- ▶ y is 0 or 1
- ▶ $\hat{y}$ will take any value between $-\infty$ and ∞
- ▶ It will be hard to find w_0 and w_1 that make the prediction $\hat{y}$ match the label y

Sigmoid Function

By applying the sigmoid function, we enforce $0 \leq \hat{y} \leq 1$

$$\hat{y} = \text{sigmoid}(w_0 + w_1 x) = \frac{1}{1 + e^{-(w_0 + w_1 x)}}$$



A new loss function

- Binary Cross Entropy Loss:

$$\text{Loss} = \frac{1}{N} \sum_{i=1}^N [-y_i \log \hat{y}_i - (1 - y_i) \log(1 - \hat{y}_i)]$$

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- What happens if $y_i = 0$?

A new loss function

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- What happens if $y_i = 0$?

$$[-y_i \log \hat{y}_i - (1 - y_i) \log(1 - \hat{y}_i)] = -\log(1 - \hat{y}_i)$$

A new loss function

- ▶ Binary Cross Entropy Loss:

$$\text{Loss} = \frac{1}{N} \sum_{i=1}^N [-y_i \log \hat{y}_i - (1 - y_i) \log(1 - \hat{y}_i)]$$

- ▶ What happens if $y_i = 0$?

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- Binary Cross Entropy Loss:

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- What happens if $y_i = 0$?

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- What happens if $y_i = 1$?

$$[-y_i \log \hat{y}_i - (1 - y_i) \log(1 - \hat{y}_i)] = -\log(\hat{y}_i)$$

MSE vs. Binary Cross Entropy Loss

- ▶ MSE of a logistic function has many local minima
- ▶ Binary Cross Entropy loss has only one minimum

Classifier

$$\hat{y} = \text{sigmoid}(w_0 + w_1 x) = \frac{1}{1 + e^{-(w_0 + w_1 x)}}$$

How to deal with uncertainty?

- Thanks to the sigmoid, $\hat{y} = f(x)$ is between 0 and 1

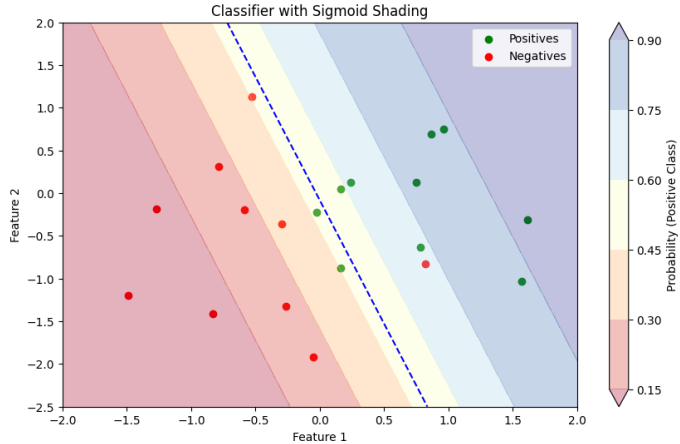
Classifier

$$\hat{y} = \text{sigmoid}(w_0 + w_1 x) = \frac{1}{1 + e^{-(w_0 + w_1 x)}}$$

How to deal with uncertainty?

- ▶ Thanks to the sigmoid, $\hat{y} = f(x)$ is between 0 and 1
- ▶ If $\hat{y}$ is close to 0, the data is probably negative
- ▶ If $\hat{y}$ is close to 1, the data is probably positive
- ▶ If $\hat{y}$ is around 0.5, we are not sure.

Classifier



Decision Boundary

- Once, we have a classifier outputting a score $0 < \hat{y} < 1$, we need to create a decision rule.

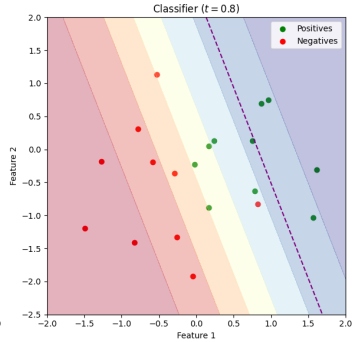
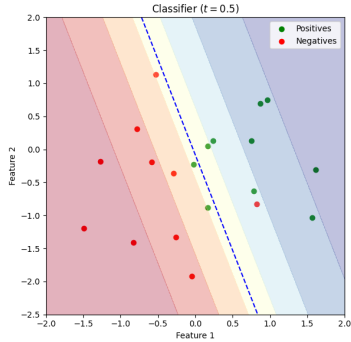
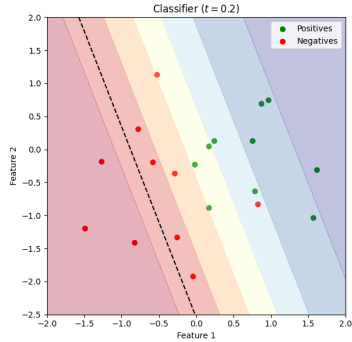
Decision Boundary

- ▶ Once, we have a classifier outputting a score $0 < \hat{y} < 1$, we need to create a decision rule.
- ▶ Let $0 < t < 1$ be a **threshold**:
 - If $\hat{y} > t$, $\hat{y}$ is classified as positive
 - If $\hat{y} < t$, $\hat{y}$ is classified as negative

Decision Boundary

- ▶ Once, we have a classifier outputting a score $0 < \hat{y} < 1$, we need to create a decision rule.
- ▶ Let $0 < t < 1$ be a **threshold**:
 - If $\hat{y} > t$, $\hat{y}$ is classified as positive
 - If $\hat{y} < t$, $\hat{y}$ is classified as negative
- ▶ How to choose t ?

Decision Boundary



Performance metrics for a classifier

- ▶ Accuracy of a classifier: percentage of correct classification
- ▶ Why accuracy alone is not a good measure for assessing the model?

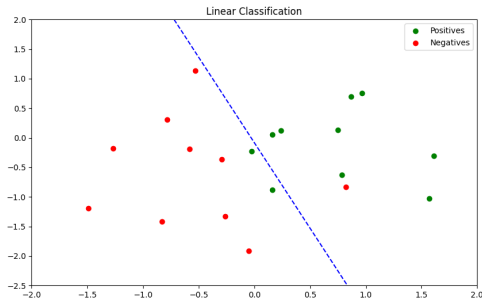
Performance metrics for a classifier

- ▶ Accuracy of a classifier: percentage of correct classification
- ▶ Why accuracy alone is not a good measure for assessing the model?
 - ▶ Example: A rare disease occurs 1 in ten thousand people
 - ▶ A test that classifies everyone as free of the disease can achieve 99.999% accuracy when tested with people drawn randomly from the entire population

Types of Errors in Classification

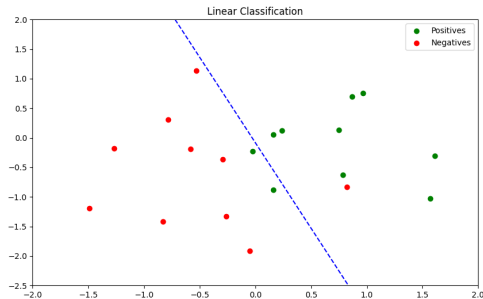
- ▶ Correct predictions:
 - True Positive (TP) : Predict $\hat{y} = 1$ when $y = 1$
 - True Negative (TN) : Predict $\hat{y} = 0$ when $y = 0$
- ▶ Two types of errors:
 - False Positive/ False Alarm (FP): $\hat{y} = 1$ when $y = 0$
 - False Negative/ Missed Detection (FN): $\hat{y} = 0$ when $y = 1$

Exercise



- ▶ How many True Positives (TP) are there?
- ▶ How many True Negatives (TN) are there?
- ▶ How many False Positives (FP) are there?
- ▶ How many False Negatives (FN) are there?

Exercise



- ▶ True Positives (TP) = 8
- ▶ True Negatives (TN) = 9
- ▶ False Positives (FP) = 1
- ▶ False Negatives (FN) = 2

Other Metrics

- Sensitivity/Recall/TPR (How many positives are detected among all positive?)

$$\frac{TP}{TP + FN}$$

- Precision (How many detected positives are actually positive?)

$$\frac{TP}{TP + FP}$$

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Diagnosing Breast Cancer

- ▶ We're going to use the breast cancer dataset to predict whether the patients' scans show a malignant tumour or a benign tumour.
- ▶ Let's try to find the best linear classifier using logistic regression.
- ▶ Open Diagnosing Breast Cancer Demo from [Course Website](#)

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Multiclass Classification

► Previous Model:

$$f(x) = \sigma(\phi(x)w)$$

► Representing Multiple Classes:

- One-hot / 1-of-K vectors, ex : 4 Class
- Class 1 : $y = [1, 0, 0, 0]$
- Class 2 : $y = [0, 1, 0, 0]$
- Class 3 : $y = [0, 0, 1, 0]$
- Class 4 : $y = [0, 0, 0, 1]$

Multiclass Classification

- ▶ Multiple outputs:

$$f(x) = \text{softmax}(\phi(x)W)$$

- ▶ Shape of $\phi(x)W$: $(N, K) = (N, D) \times (D, K)$
- ▶ Softmax:

$$\text{softmax}(z_k) = \frac{e^{z_k}}{\sum_j e^{z_j}}$$

Softmax Example

$$z = \begin{bmatrix} -1 \\ 2 \\ 1 \\ -4 \end{bmatrix}$$

$$\text{softmax}(z) = \begin{bmatrix} \frac{e^{-1}}{e^{-1}+e^2+e^1+e^{-4}} \\ \frac{e^2}{e^{-1}+e^2+e^1+e^{-4}} \\ \frac{e^1}{e^{-1}+e^2+e^1+e^{-4}} \\ \frac{e^{-4}}{e^{-1}+e^2+e^1+e^{-4}} \end{bmatrix} \approx \begin{bmatrix} 0.035 \\ 0.704 \\ 0.259 \\ 0.002 \end{bmatrix}$$

Cross-Entropy

- ▶ Multiple Outputs: $\hat{y}_i = \text{softmax}(\phi(x_i)W)$
- ▶ Cross-Entropy:

$$J(W) = - \sum_{i=1}^N \sum_{k=1}^K y_{ik} \log(\hat{y}_{ik})$$

- ▶ Example, $K = 4$, if $y_i = [0, 0, 1, 0]$ then,

$$\sum_{k=1}^K y_{ik} \log(\hat{y}_{ik}) = \log(\hat{y}_{i3})$$

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Iris Dataset

- Open Iris Dataset Demo from [Course Website](#)